

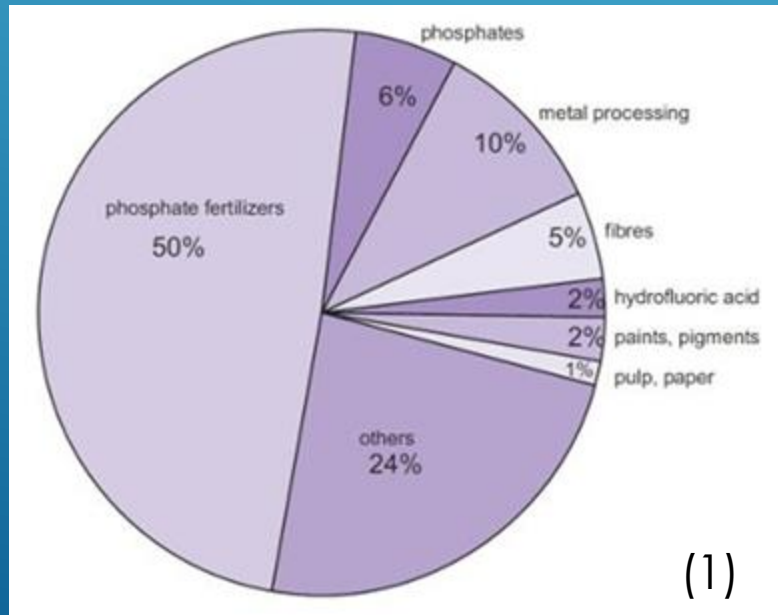


OPTIMIZED PROCESS FOR SULFURIC ACID PRODUCTION

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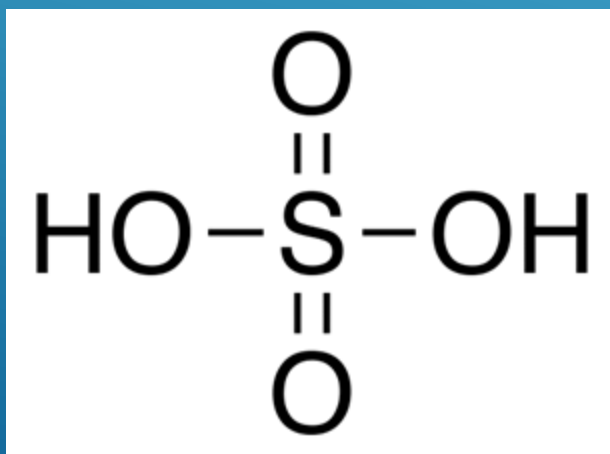
INTRODUCTION

- Sulfuric acid, also known by medieval alchemists as 'oil of vitriol', is one of the most largely produced chemicals in the world, and for the United States, it holds the number one spot (1).
- Sulfuric acid has a myriad of uses, which include but is not limited to, phosphate fertilizers, oil refining, motor vehicle batteries, metal processing, dyes, and detergents.





- Sulfuric acid is a mineral acid composed of sulfur, oxygen and hydrogen. It is colorless, odorless, and has a syrupy consistency.
- It is considered a strong acid, and therefore a strong oxidizer. It also has dehydrating properties and is hygroscopic.



MARKET ANALYSIS

- Worldwide, about 200+ million tons of sulfuric acid are produced each year, with an annual growth rate of about 3% due to high demand from Asia (4).
- Why the price spike in 2008?
- This was due to the inability of the world suppliers to meet the demand, an inflexible supply caused the price skyrocket

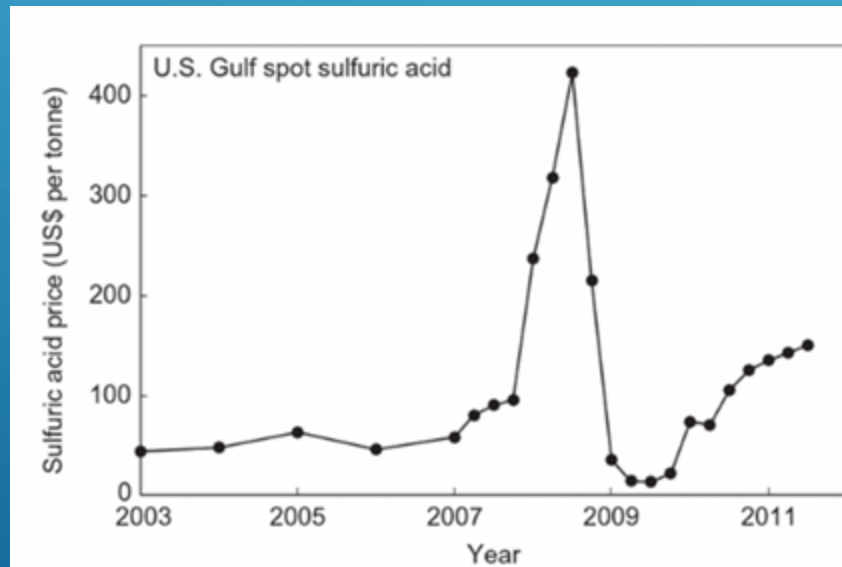


Figure 2: U.S. Gulf port pricing, showing the spike in 2009 (4)

Customer Requirements

- The supplier companies need to deliver on two main needs: concentration and storage.
- Pure, high strength acid is needed for most of the products mentioned in the introduction, including metal, dye, and fertilizer production. This concentration is generally anywhere from 93% to 99.99% pure.
- It is an aggressive acid which can corrode metal and even oxidize plastics. Not only is it corrosive, but also very heavy. At 93-99.99% concentration, sulfuric acid is almost twice the weight of water, and needs appropriate storage containers (3).

Problem

- Manufacturing of sulfuric acid is crucial for many industries
- Can be manufacture by two processes:
 - Lead Chamber Process
 - Contact Process

Lead Chamber Process

- Earliest design process for production of H_2SO_4 using sulfur dioxide and nitric acid.
- 3 main towers:
 - Glover Tower: SO_2 to SO_3
 - Lead Chambers: H_2SO_4 condense on the wall and collected in bottom
 - Gay-Lussac: unreacted gases treated with cool conc. H_2SO_4 and fed back to Glover Tower
- Produced ~70% of the acid and is no longer in use.

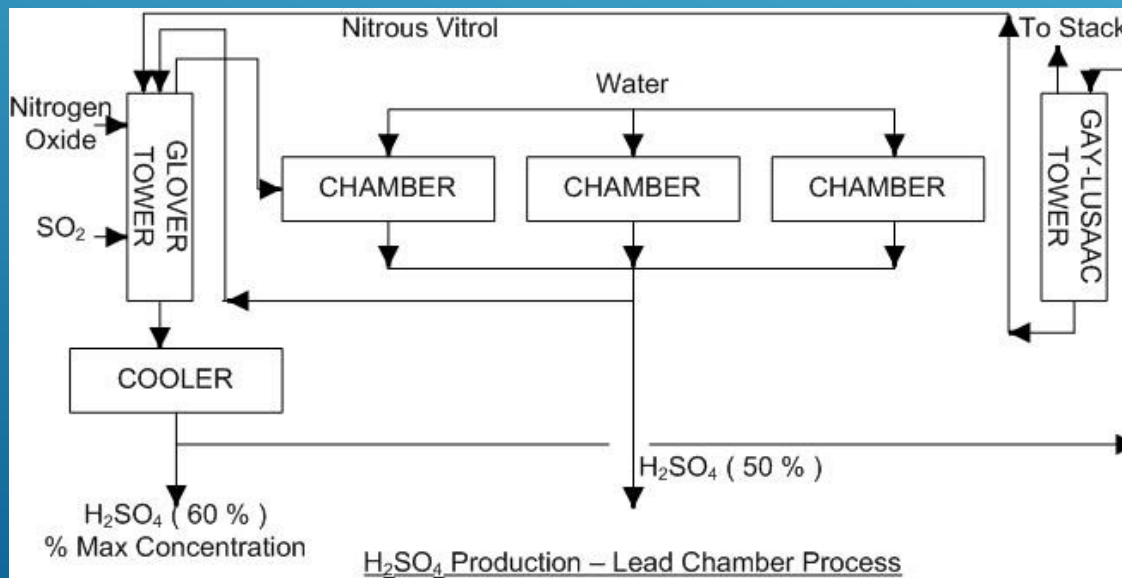


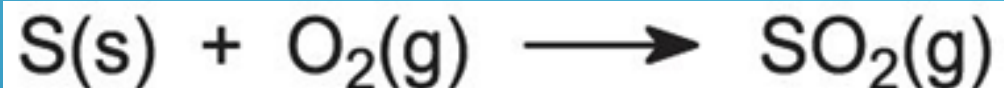
Figure 3: BFD of Lead Chamber Process(12)

Contact Process

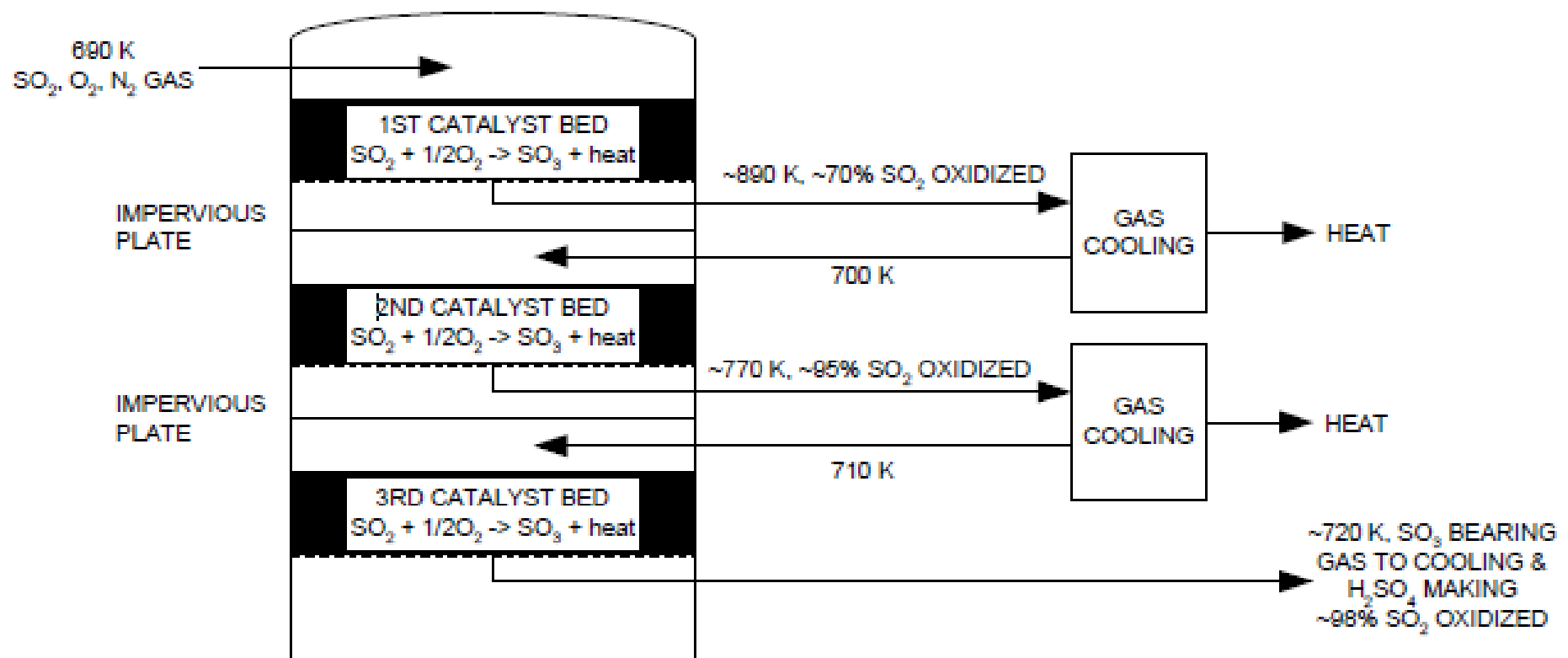
- Molten Sulfur burnt in presence of dry air to produce SO_2
- Fed to fixed bed reactor with 4 catalytic (V_2O_5) beds with intermediate gas cooling to oxidize sulfur dioxide to sulfur trioxide.
- The next step varies for different types of process i.e.
 - Single Contact
 - Double Contact
 - Wet Sulfuric Acid (WSA)
- The sulfur dioxide stream gets absorbed in absorber tower with a recycle feed of sulfuric acid to form ~99 - 99.99% of sulfuric acid.

Contact Process

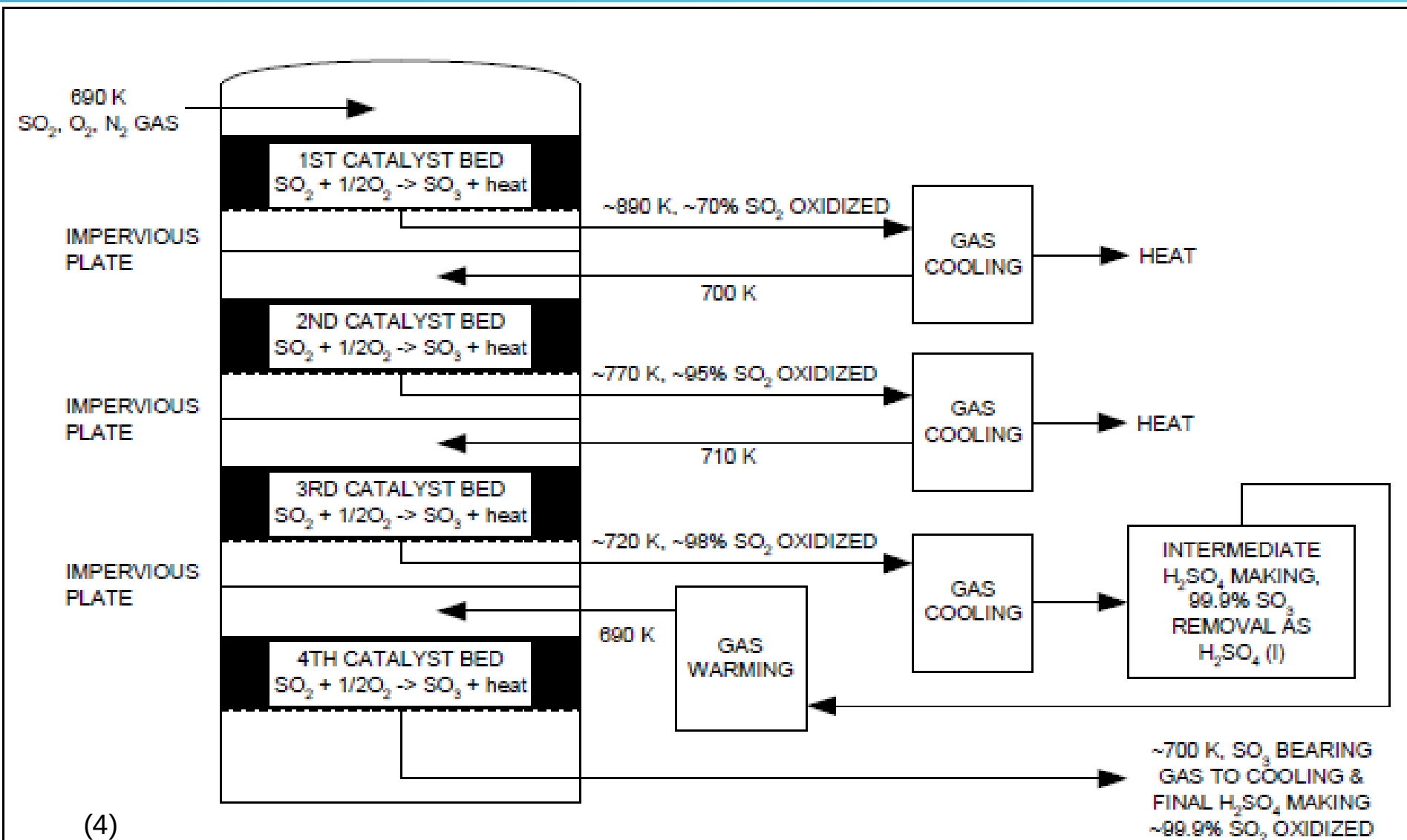
Here are the general reactions for the contact process:



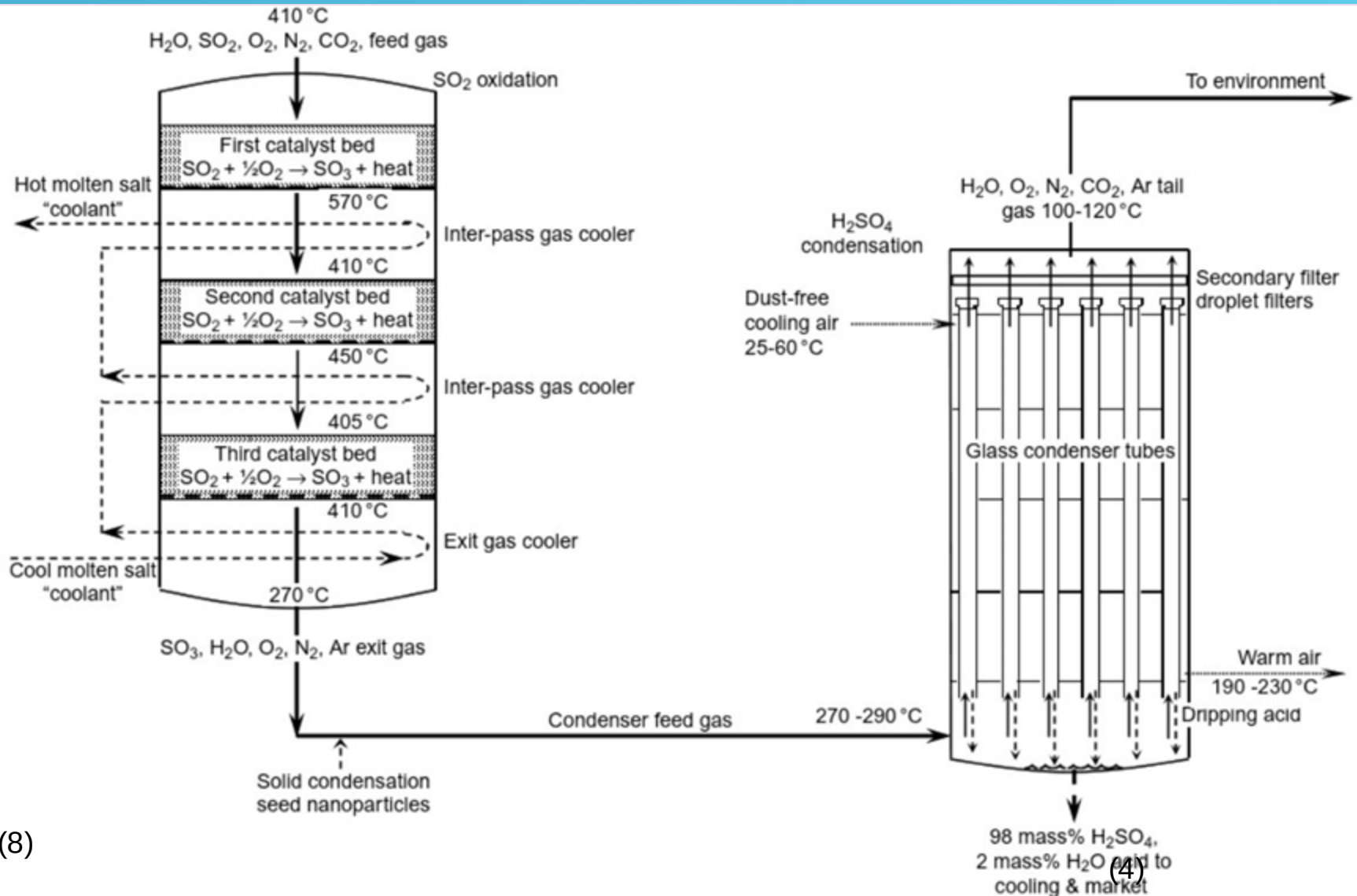
Single Contact Method



Double Contact Method



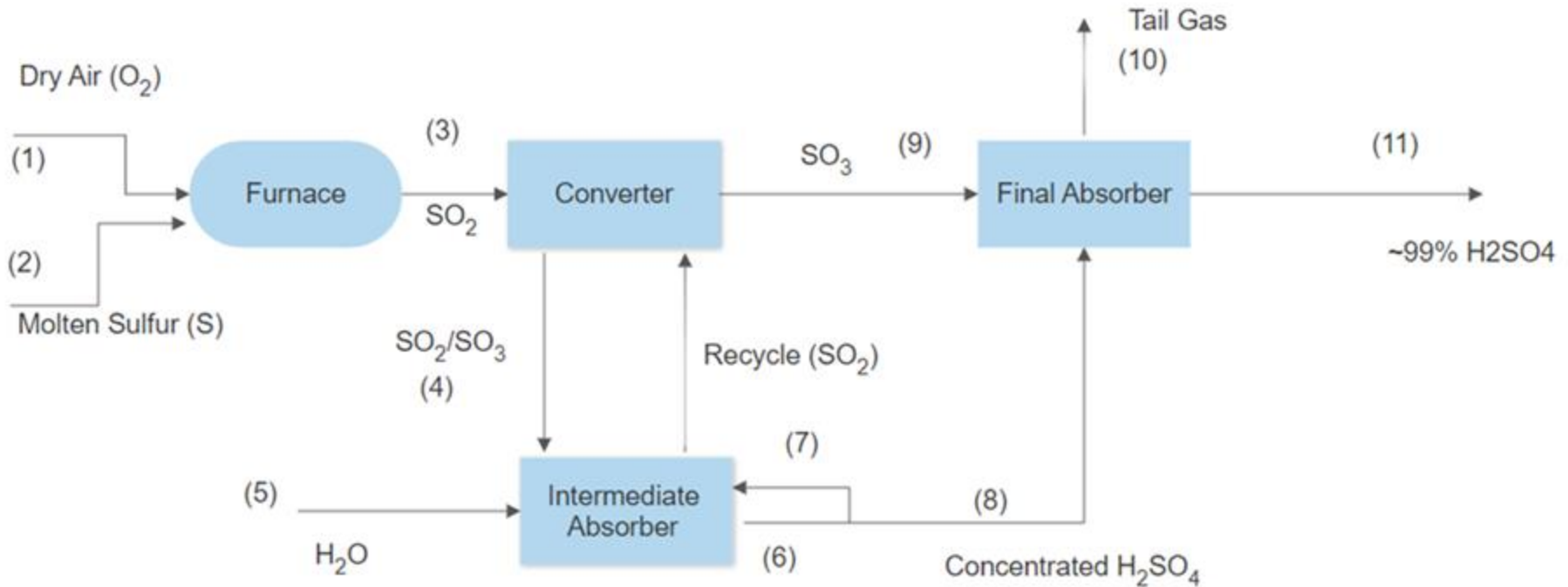
Wet Gas Sulfuric Acid (WSA) Method



DESIGN BASIS

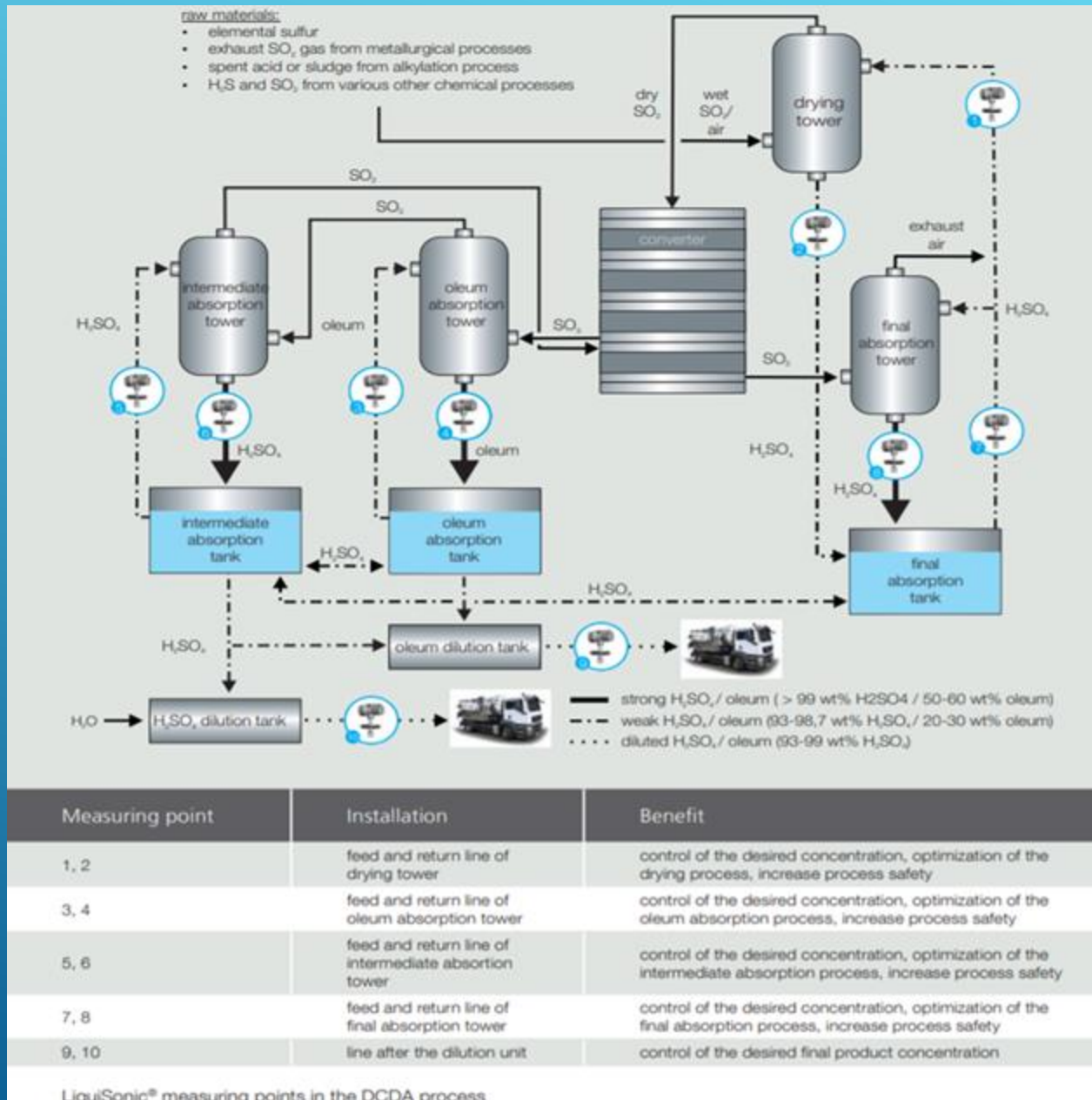
- The double contact process chosen as the solution in this presentation will produce a ~99.9% weight sulfuric acid product.
- The design basis for this report aims to produce 5% of the 3% worldwide growth. This percentage calculates to a 300,000 ton/year with a 330-day work week
- $300,000 \frac{\text{tons}}{\text{year}} * \frac{\text{year}}{330 \text{ day}} * \frac{1 \text{ day}}{24 \text{ hours}} * \frac{2000 \text{ lbs}}{1 \text{ ton}} = 75,757 \frac{\text{lb}}{\text{hr}}$
- The units used for this process will be from the English system, as demonstrated in the above calculation. This is done to be consistent with the market standard.
- The location of the plant will be in the Midwest of the United States. This location was chosen based on the proximity to the companies who produce phosphate fertilizers.

BFD



BLOCK FLOW DIAGRAM

PFD



MATERIAL BALANCES

Stream No.	(1)	(2)	(3)	(11)
Stream Components	Dry Air (O ₂)	Molten Sulfur (S)	Sulfur dioxide with other gases	Product acid
S	-	6.926	-	-
O ₂	1.45152	-	0.699	-
SO ₂	-	-	1.52	-
SO ₃	-	-	-	-
H ₂ SO ₄	-	-	-	2.14
Total	1.45152	6.926	2.219	2.14

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